Chapter 5

Convergence

The convergence of sequences of random variables to some limit random variable is an important concept in probability theory, and its applications to statistics and stochastic processes. The same concepts are known in more general mathematics as stochastic convergence. There exist several modes of convergence of random variables, some of which are described below.

5.1 Convergence in Distribution

With this mode of convergence, we increasingly expect to see the next outcome in a sequence of random experiments becoming better and better modeled by a given probability distribution. Convergence in distribution is the weakest form of convergence. However, convergence in distribution is very frequently used in practice; most often it arises from application of the central limit theorem.

5.1.1 Definition

A sequence of real valued random variables $\{X_n\}$, $n = 1, 2, \cdots$ is said to be **converge in distribution**, or **converge weakly**, or **converge in law** to a random variable X if it follows as

$$\lim_{n\to\infty} F_n(x) = F(x),$$

for every number $x \in \mathbb{R}$ at which F is continuous. Here F_n and F are the cumulative distribution functions of random variables X_n and X, respectively.

Convergence in distribution may be denoted as

$$X_n \xrightarrow{L} X \text{ or } X_n \xrightarrow{d} X.$$

5.1.2 Properties

Some of the properties of convergence in distribution are as follow:

1. Let $X_n \xrightarrow{L} X$ and c is a constant, then

   (i) $X_n + c \xrightarrow{L} X + c$, and

   (ii) $cX_n \xrightarrow{L} cX$.

2. $X_n \xrightarrow{L} X$ and $Y_n \xrightarrow{L} c$, then

   (i) $X_n + Y_n \xrightarrow{L} X + c$,

   (ii) $X_n Y_n \xrightarrow{L} cX$, and

   (iii) $\frac{X_n}{Y_n} \xrightarrow{L} \frac{X}{c}$ $(c \neq 0)$.

5.1.3 Examples

Example 1: Let $\{X_n\}$ be a sequence of binomial variates having distribution

$$Pr\left[X_n = r\right] = \binom{n}{r} {P_n}^r \left(1 - P_n\right)^{n-r}.$$

Let $E\left[X_n\right] = nP_n = \lambda$ be a finite constant, then

$$Pr\left[X_n = r\right] \to \frac{e^{-\lambda}\lambda^r}{r!} \text{ as } n \to \infty.$$

Example 2: Let $\{X_n\}$ be a sequence of t variates having distribution

$$f(X_n = t) = \frac{1}{\sqrt{n}\, B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}.$$

then it follows that

$$f(X_n = t) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}t^2} \text{ as } n \to \infty.$$

5.2 Convergence in Probability

The basic idea behind this type of convergence is that the probability of an "unusual" outcome becomes smaller and smaller as the sequence progresses. The concept of convergence in probability is used very often in statistics. *For example*, an estimator is called consistent if it converges in probability to the quantity being estimated. Convergence in probability is also the type of convergence established by the weak law of large numbers.

5.2.1 Definition

A sequence of random variables $\{X_n\}$, $n = 1, 2, \cdots$ is said to be **converges in probability** towards the random variable X if for all $\epsilon > 0$

$$\lim_{n\to\infty} Pr\left(|X_n - X| > \epsilon\right) = 0.$$

Convergence in probability is denoted by adding the letter p over an arrow indicating convergence, or using the"plim" probability limit operator:

$$X_n \xrightarrow{p} X \text{ or } X_n \xrightarrow{P} X \text{ or } \operatorname*{plim}_{n\to\infty} X_n = X.$$

5.2.2 Properties

Some of the properties of convergence in probability are as follow:

1. $X_n \xrightarrow{P} X \quad \Rightarrow X_n - X \xrightarrow{P} 0.$

2. $X_n \xrightarrow{P} X,\ X_n \xrightarrow{P} Y \quad \Rightarrow Pr[X = Y] = 1.$

3. $X_n \xrightarrow{P} X,\ Y_n \xrightarrow{P} Y$, then

(i) $X_n \pm Y_n \xrightarrow{P} X \pm Y$

(ii) $X_n Y_n \xrightarrow{P} XY$

(iii) $\frac{X_n}{Y_n} \xrightarrow{P} \frac{X}{Y}$

4. $X_n \xrightarrow{P} X$, and k is a constant $\Rightarrow kX_n \xrightarrow{P} kX$.

5. $X_n \xrightarrow{P} k \quad \Rightarrow X_n^2 \xrightarrow{P} k^2$.

6. $X_n \xrightarrow{P} a,\ Y_n \xrightarrow{P} b \quad \Rightarrow X_n Y_n \xrightarrow{P} ab$, where a, b are constants.

7. $X_n \xrightarrow{P} X$ and Y is a random variable $\Rightarrow X_n Y \xrightarrow{P} XY$.

8. $X_n \xrightarrow{P} 1 \quad \Rightarrow X_n^{-1} \xrightarrow{P} 1$.

9. Convergence in probability implies convergence in distribution: $X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{d} X$.

All the above statement can be easily varified.

5.2.3 Some Examples

Example 1: Let $X_1, X_2, \cdots$ be a sequence of independent and identically distributed random variables with $\mu = E(X_i)$ and $\sigma^2 = Var(X_i) < \infty$ for $i = 1, 2, \cdots, \infty$. Then sample mean $\bar{X}_n$ converges in probability to μ.

Solution: We have the following

$$E\left(\bar{X}_n\right) = \mu \quad \text{and} \quad Var\left(\bar{X}_n\right) = \frac{\sigma^2}{n}.$$

By Chebyshev's inequality

$$Pr\left(\left|\bar{X}_n - E\left(\bar{X}_n\right)\right| \geq \epsilon\right) \leq \frac{Var\left(\bar{X}_n\right)}{\epsilon^2},$$

for $\epsilon > 0$. Hence

$$Pr\left(\left|\bar{X}_n - \mu\right| \geq \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2}.$$

Taking the limit as n tends to infinity, we get

$$\lim_{n\to\infty} Pr\left(|\bar{X}_n - \mu| \geq \epsilon\right) \leq \lim_{n\to\infty} \frac{\sigma^2}{n\epsilon^2},$$

which yields

$$\lim_{n\to\infty} Pr\left(|\bar{X}_n - \mu| \geq \epsilon\right) = 0.$$

So the sample mean $\bar{X}_n$ converges in probability to μ, also written as $\bar{X}_n \xrightarrow{P} \mu$.

Example 2: Let $X_n \sim Exponential(n)$, show that $X_n \xrightarrow{P} 0$. That is, the sequence $X_1, X_2, \cdots$ converges in probability to the zero random variable X.

Solution: We have

$$\begin{aligned}
\lim_{n\to\infty} Pr\left(|X_n - 0| \geq \epsilon\right) &= \lim_{n\to\infty} Pr\left(X_n \geq \epsilon\right) \\
&= \lim_{n\to\infty} \left[1 - Pr\left(X_n \leq \epsilon\right)\right] \\
&= \lim_{n\to\infty} \left[1 - \left(1 - e^{-n\epsilon}\right)\right] \\
&= \lim_{n\to\infty} e^{-n\epsilon} \\
&= 0, \quad \text{for all } \epsilon > 0.
\end{aligned}$$

Hence the sequence of random variables $X_1, X_2, \cdots$ convergence in probability to the zero random variable X.

Example 3: If $X \sim Binomial(n, p)$, then $X_n \xrightarrow{P} p$.

Solution: **Assignment**

Theorem 5.2.1. *Convergence in probability implies convergence in distribution. That can be also written as*

$$X_n \xrightarrow{P} X \quad \Rightarrow X_n \xrightarrow{d} X.$$

Proof. In order to prove convergence in distribution, one must show that the sequence of cumulative distribution functions converges to the F_X at every point where F_X is continuous. Let a be such a point. For every $\epsilon > 0$, we have

$$\Pr(X_n \leq a) \leq \Pr(X \leq a + \varepsilon) + \Pr(|X_n - X| > \varepsilon)$$
$$\Pr(X \leq a - \varepsilon) \leq \Pr(X_n \leq a) + \Pr(|X_n - X| > \varepsilon)$$

So, we have

$$\begin{aligned} \Pr(X \leq a - \varepsilon) - \Pr\left(|X_n - X| > \varepsilon\right) &\leq \Pr(X_n \leq a) \\ &\leq \Pr(X \leq a + \varepsilon) + \Pr\left(|X_n - X| > \varepsilon\right). \end{aligned}$$

Taking the limit as $n \to \infty$, we obtain:

$$F_X(a - \varepsilon) \leq \lim_{n\to\infty} \Pr(X_n \leq a) \leq F_X(a + \varepsilon),$$

where $F_X = Pr(X \leq a)$ is the cumulative distribution function of X. This function is continuous at a by assumption, and therefore both $F_X(a - \epsilon)$ and $F_X(a + \epsilon)$ converge to $F_X(a)$ as $\epsilon \to 0^+$. Taking this limit, we obtain

$$\lim_{n\to\infty} \Pr(X_n \leq a) = \Pr(X \leq a),$$

which means that $\{X_n\}$ converges to X in distribution. □

5.3 Almost Sure Convergence

This is the type of stochastic convergence that is most similar to pointwise convergence known from elementary real analysis.

5.3.1 Definition

A sequence of random variables $\{X_n\}$, $n = 1, 2, \cdots$ is said to be converges **almost surely (a.s)** or **almost everywhere** or **with probability 1** or **strongly** to another random variable X if and only if

$$Pr\left[\lim_{n\to\infty} X_n = X\right] = 1,$$

or, equivalently,

$$\lim_{N\to\infty} Pr\left[\sup_{n\geq N} |X_n - X| > \epsilon\right] = 0 \text{ for every } \epsilon.$$

Almost sure convergence is often denoted by adding the letters $a.s.$ over an arrow indicating convergence: $X_n \stackrel{a.s.}{\to} X$ or $X_n \to X$ with probability 1.

5.3.2 Properties

Some of the properties of almose surely are as follow:

1. If $X_n \stackrel{a.s.}{\to} X$ and $X_n \stackrel{a.s.}{\to} Y$, then $X = Y$ almost surely.
2. If $X_n \stackrel{a.s.}{\to} X$ and $Y_n \stackrel{a.s.}{\to} Y$, then $aX_n + bY_n \stackrel{a.s.}{\to} aX + bY,\ \forall\ a, b \in \mathbb{R}$.
3. If $X_n \stackrel{a.s.}{\to} X$ and $Y_n \stackrel{a.s.}{\to} Y$, then $X_nY_n \stackrel{a.s.}{\to} XY$.
4. Almost sure convergence implies convergence in probability, and hence implies convergence in distribution: $X_n \stackrel{a.s.}{\to} X \ \Rightarrow X_n \stackrel{P}{\to} X \ \Rightarrow X_n \stackrel{d}{\to} X$.
5. Convergence in probability implies there exists a sub-sequence (k_n) which almost surely converges: $X_n \stackrel{P}{\to} X \ \Rightarrow X_{k_n} \stackrel{a.s.}{\to} X$.

5.3.3 Examples

Example 1: Consider an animal of some short-lived species. We record the amount of food that this animal consumes per day. This sequence of numbers will be unpredictable, but we may be quite certain that one day the number will become zero, and will stay zero forever after.

Example 2: Consider a man who tosses seven coins every morning. Each afternoon, he donates one pound to a charity for each head that appeared. The first time the result is all tails, however, he will stop permanently.

Let $X_1, X_2, \cdots$ be the daily amounts the charity received from him. We may be almost sure that one day this amount will be zero, and stay zero forever after that.

Theorem 5.3.1. *Convergence almost surely implies convergence in probability. That can be also written as*

$$X_n \stackrel{a.s.}{\to} X \quad \Rightarrow X_n \stackrel{P}{\to} X.$$

Proof. We have

$$|X_n - X| > \epsilon \quad \Rightarrow \sup_{n \geq N} |X_n - X| > \epsilon.$$

So that

$$Pr\left[|X_n - X| > \epsilon\right] \leq Pr\left[\sup_{n \geq N} |X_n - X| > \epsilon\right].$$

According to the law of large numbers

$$Pr\left[\sup_{n \geq N} |X_n - X| > \epsilon\right] = 0 \text{ as } n \to \infty.$$

Therefore,

$$Pr\left[|X_n - X| > \epsilon\right] \to 0 \text{ as } n \to \infty.$$

Hence, $X_n \stackrel{a.s.}{\to} X \quad \Rightarrow X_n \stackrel{P}{\to} X.$ □

5.4 Laws of Large Numbers

The law of large numbers (LLN) is a theorem that describes the result of performing the same experiment a large number of times. According to the law, the average of the results obtained from a large number of trials should be close to the expected value, and will tend to become closer as more trials are performed.

The LLN is important because it guarantees stable long-term results for the averages of some random events.

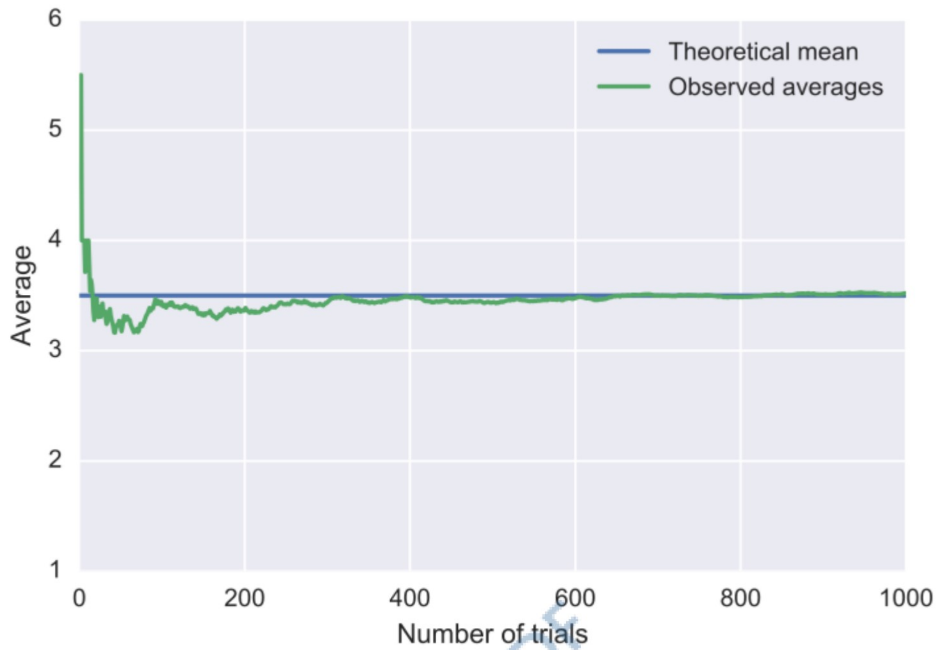

Figure 5.1: Average Dice Roll by Number of Rolls.

An illustration of the law of large numbers using a particular run of rolls of a single dice. As the number of rolls in this run increases, the average of the values of all the results approaches 3.5. While different runs would show a different shape over a small number of throws (at the left), over a large number of rolls (to the right) they would be extremely similar.

Two different versions of the law of large numbers are described below; they are called the **strong law of large numbers**, and the **weak law of large numbers**.

Chebyshev's Inequality: Let X be a random variable with mean μ and standard deviation σ. Then Chebyshev's inequality states that

$$P(|X-\mu|) \geq k\sigma) \leq \frac{1}{k^2} \qquad \text{Or,} \quad P(|X-\mu|) < k\sigma) \geq 1-\frac{1}{k^2},$$

for any nonzero positive constant k.

5.4.1 Weak Law of Large Numbers

The weak law of large numbers (WLLN) states that the sample average converges in probability towards the expected value

$$\overline{X}_n \xrightarrow{P} \mu, \qquad \text{when } n \to \infty.$$

That is, for any positive number ϵ,

$$\lim_{n\to\infty} \Pr\big(\,|\overline{X}_n - \mu| > \varepsilon\,\big) = 0.$$

Interpreting this result, the weak law states that for any nonzero margin specified, no matter how small, with a sufficiently large sample there will be a very high probability that the average of the observations will be close to the expected value; that is, within the margin.

Necessary and Sufficient Conditions

Necessary and sufficient conditions for the existence of WLLN are

(i) $E\left(X_i\right)$ exists for all i,

(ii) $B_n = Var\left[X_1, X_2, \cdots, X_n\right]$ exists and

(iii) $\frac{B_n}{n^2} \to 0$ as $n \to \infty$.

Condition (i) is necessary without it the law itself cannot be stated. But the condition (ii) and (iii) are not necessary, (iii) is however a sufficient condition.

Theorem 5.4.1 (Chebyshev's Law of Large Numbers)**.** *Given $X_1, X_2, \cdots$ an infinite sequence of i.i.d. random variables with finite expected value $E\left(X_i\right) = \mu < \infty\ \forall\ i$, we are interested in the convergence of the sample average $\overline{X}_n = \frac{1}{n}(X_1 + \cdots + X_n)$.*

The weak law of large numbers states:

$$\overline{X}_n \xrightarrow{P} \mu \qquad \textit{when } n \to \infty.$$

Proof. This proof uses the assumption of finite $Var(X_i) = \sigma^2$ (for all i). The independence of the random variables implies no correlation between them, and we have that as the following

$$\begin{aligned} Var(\overline{X}_n) &= \text{Var}(\tfrac{1}{n}(X_1 + \cdots + X_n)) \\ &= \frac{1}{n^2}\text{Var}(X_1 + \cdots + X_n) \\ &= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}. \end{aligned}$$

The common mean μ of the sequence is the mean of the sample average:

$$E(\overline{X}_n) = \mu.$$

Using Chebyshev's inequality on $\overline{X}_n$ results in

$$\text{P}(\left|\overline{X}_n - \mu\right| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

This may be used to obtain the following:

$$\text{P}(\left|\overline{X}_n - \mu\right| < \varepsilon) = 1 - \text{P}(\left|\overline{X}_n - \mu\right| \geqslant \varepsilon) \geqslant 1 - \frac{\sigma^2}{n\varepsilon^2}.$$

As n approaches infinity, the expression approaches 1. And by definition of convergence in probability, we have obtained

$$\overline{X}_n \xrightarrow{P} \mu \qquad \text{when } n \to \infty.$$

Hence the theorem. □

5.4.2 Strong Law of Large Numbers

The strong law of large numbers states that the sample average converges almost surely to the expected value

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu, \qquad \text{when } n \to \infty.$$

That is,

$$\Pr\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1.$$

What this means is that the probability that, as the number of trials "n" goes to infinity, the average of the observations converges to the expected value, is equal to one.

Necessary and Sufficient Conditions

Necessary and sufficient conditions for the existence of SLLN are

(i) $E(X_i)$ exists for all i, and

(ii) $\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} < \infty$.

Condition (i) is necessary for the existence of strong law of large numbers and condition (ii) is sufficient. That is convergence of $\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2}$ is sufficient for the existence of strong law of large numbers.

Theorem 5.4.2 (Kolmogorov Law of Large Numbers)**.** *Let* $\{X_i\}, \ i = 1, 2, \cdots,$ *be a sequence of independent random variables such that* $E(X_i) = \mu_i$ *and* $Var(X_i) = \sigma_i^2$*. Then hold the following*

$$\sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} < \infty \quad \Rightarrow \bar{X}_n - \bar{\mu}_n \xrightarrow{a.s.} 0,$$

that is, the sequence $X_1, X_2, \cdots$ *obeys the strong law of large numbers.*

Proof. **Assignment** ☐

Assignment: *Differentiate between WLLN and SLLN.*

Example: Let X_i assume two values i and $-i$ with equal probabilities. Show that the law of large numbers cannot be applied to the independent random variables $X_1, X_2, \cdots$, i.e., $X's$.

Solution: We have

$$P\left[X_i = i\right] = \frac{1}{2}, \; P\left[X_i = -i\right] = \frac{1}{2}.$$

So we can obtain as

$$E\left[X_i\right] = \frac{1}{2}(i) + \frac{1}{2}(-i) = 0, \; i = 1, 2, \cdots,$$

and

$$Var\left[X_i\right] = E\left[X_i^2\right] = \frac{i^2}{2} + \frac{i^2}{2} = i^2, \; i = 1, 2, \cdots.$$

Since $X_1, X_2, \cdots, X_n$ are independent random variables, we can obtain as

$$\begin{aligned}
\frac{B_n}{n^2} &= \frac{1}{n^2} Var\left[X_1 + X_2 + \cdots + X_n\right] \\
&= \frac{1}{n^2}\left[1^2 + 2^2 + \cdots + n^2\right] \\
&= \frac{n(n+1)(2n+1)}{6n^2} \\
&= \frac{n(1+\frac{1}{n})(2+\frac{1}{n})}{6} \to \infty \text{ as } n \to \infty.
\end{aligned}$$

Hence law of large numbers does not hold.

Example: Let X_i can have two values i^α and $-i^\alpha$ with equal probabilities. Show that the law of large numbers can be applied to the independent random variables $X_1, X_2, \cdots$, if $\alpha < \frac{1}{2}$.

Solution: We have

$$P[X_i = i^\alpha] = \frac{1}{2},\ P[X_i = -i^\alpha] = \frac{1}{2}.$$

So we can obtain as

$$E[X_i] = \frac{1}{2}(i^\alpha) + \frac{1}{2}(-i^\alpha) = 0,\ i = 1, 2, \cdots,$$

and

$$Var[X_i] = E\left[X_i^2\right] = \frac{(i^\alpha)^2}{2} + \frac{(-i^\alpha)^2}{2} = i^{2\alpha},\ i = 1, 2, \cdots.$$

Since $X_1, X_2, \cdots, X_n$ are independent random variables, we can obtain as

$$\begin{aligned}
\frac{B_n}{n^2} &= \frac{1}{n^2} Var[X_1 + X_2 + \cdots + X_n] \\
&= \frac{1}{n^2}\left[1^{2\alpha} + 2^{2\alpha} + \cdots + n^{2\alpha}\right] \\
&= \frac{1}{n^2}\int_0^n x^{2\alpha}\, dx \qquad \text{[From Euler-Maclaurin's Formula]} \\
&= \frac{1}{n^2}\left[\frac{x^{2\alpha+1}}{2\alpha+1}\right]_0^n \\
&= \frac{1}{n^2} \cdot \frac{n^{2\alpha+1}}{2\alpha+1} \\
&= \frac{n^{2\alpha-1}}{2\alpha+1} \to 0 \text{ as } n \to \infty \text{ if } \alpha < \frac{1}{2}.
\end{aligned}$$

Hence the results follows law of large numbers.

Example: Let $\{X_n\}$ be mutually independent and identically distributed random variables with mean μ and finite variance. If $S_n = X_1 + X_2 + \cdots + X_n$, prove that the low of large numbers does not hold for the sequence $\{S_n\}$.

Solution: Since $S_1, S_2, \cdots, S_n$ are mutually independent and identically distributed random variables, so

$$\begin{aligned}
\frac{B_n}{n^2} &= \frac{1}{n^2} Var\left[S_1 + S_2 + \cdots + S_n\right] \\
&= \frac{1}{n^2} Var\left[X_1 + (X_1 + X_2) + \cdots + (X_1 + X_2 + \cdots + X_n)\right] \\
&= \frac{1}{n^2} Var\left[nX_1 + (n-1)X_2 + \cdots + 2X_{n-1} + X_n\right] \\
&= \frac{1}{n^2}\left[n^2 Var\left(X_1\right) + (n-1)^2 Var\left(X_2\right) + \cdots + 2^2 Var\left(X_{n-1}\right) + 1^2 Var\left(X_n\right)\right]
\end{aligned}$$

Let $Var\left(X_i\right) = \sigma^2$ for all i, therefore

$$\begin{aligned}
\frac{B_n}{n^2} &= \frac{\sigma^2}{n^2}\left[1^2 + 2^2 + \cdots + n^2\right] \\
&= \frac{\sigma^2 n(n+1)(2n+1)}{6n^2} \\
&= \frac{\sigma^2 n(1+\frac{1}{n})(2+\frac{1}{n})}{6} \to \infty \text{ as } n \to \infty.
\end{aligned}$$

Hence law of large numbers does not hold for the sequence $\{S_n\}$.

5.5 Central Limit Theorem

The central limit theorem (CLT) establishes that, in some situations, when independent random variables are added, their properly normalized sum tends toward a normal distribution (informally a "bell curve" distribution) even if the original variables themselves are not normally distributed. The theorem is a key concept in probability theory because it implies that probabilistic and statistical methods that work for normal distributions can be applicable to many problems involving other types of distributions.

For example, suppose that a sample is obtained containing a large number of observations, each observation being randomly generated in a way that does not depend on the values of the other observations, and that the arithmetic mean of the observed values is computed. If this procedure is performed many times, the central limit theorem says that the distribution of the average will be closely approximated by a normal distribution. A simple example of this is that if one flips a coin many times the probability of getting a given number of heads in a series of flips will approach a normal curve, with mean equal to half the total number of flips in each series.

This theorem was first stated by Laplace in 1812 and a regorous proof under fairly general conditions was given by Liapounoff in 1901. Below list some particular cases of this general central limit theorem.

(i) De-Moirve's Laplace theorem,

(ii) Lindeberge-Levy theorem,

(iii) Liapounov's theorem, and

(iv) Lindeberg-Feller theorem.

5.5.1 Classical Central Limit Theorem

Let $\{X_1, X_2, \cdots, X_n\}$ be a random sample of size n, that is, a sequence of independent and identically distributed random variables drawn from a distribution of expected value given by μ and finite variance given by σ^2. Suppose we are interested in the sample average

$$S_n = \frac{X_1 + \cdots + X_n}{n}$$

of these random variables. By the law of large numbers, the sample averages converge in probability and almost surely to the expected value μ as $n \to \infty$. The classical central limit theorem describes the size and the distributional form of the stochastic fluctuations around the deterministic number μ during this convergence. More precisely, it states that as n gets larger, the distribution of the difference between the sample average S_n and its limit μ, when multiplied by the factor $\sqrt{n}$ (that is $\sqrt{n}\,(S_n - \mu)$), approximates the normal distribution with mean 0 and variance σ^2. For large enough n, the distribution of S_n is close to the normal distribution with mean μ and variance

$\frac{\sigma^2}{n}$. The usefulness of the theorem is that the distribution of $\sqrt{n}\,(S_n - \mu)$ approaches normality regardless of the shape of the distribution of the individual X_i. Formally, the theorem can be stated as follows:

Theorem 5.5.1 (Lindeberg–Lévy CLT). *Suppose $\{X_1, X_2, \cdots\}$ is a sequence of independent and identically distributed random variables with $E\,(X_i) = \mu$ and $Var\,(X_i) = \sigma^2 < \infty$. Then as n approaches infinity, the random variables $\sqrt{n}\,(S_n - \mu)$ convergence in distribution to a normal $N\,(0, \sigma^2)$:*

$$\sqrt{n}\,(S_n - \mu) \ \xrightarrow{d} \ N\left(0, \sigma^2\right).$$

Proof. The central limit theorem has a simple proof using characteristic function. It is similar to the proof of the (weak) law of large numbers.

Assume $\{X_1, X_2, \cdots, X_n\}$ are independent and identically distributed random variables, each with mean μ and finite variance σ^2. The sum $X_1 + X_2 + \cdots + X_n$ has mean $n\mu$ and variance $n\sigma^2$. Consider the random variable

$$Z_n \ = \ \frac{X_1 + \cdots + X_n - n\mu}{\sqrt{n\sigma^2}} \ = \ \sum_{i=1}^{n} \frac{X_i - \mu}{\sqrt{n\sigma^2}} \ = \ \sum_{i=1}^{n} \frac{1}{\sqrt{n}} Y_i,$$

where in the last step we defined the new random variables $Y_i = \frac{X_i - \mu}{\sigma}$, each with zero mean and unit variance. The characteristic function of Z_n is given by

$$\varphi_{Z_n}(t) \ = \ \varphi_{\sum_{i=1}^{n} \frac{1}{\sqrt{n}} Y_i}(t) \ = \ \varphi_{Y_1}\!\left(\frac{t}{\sqrt{n}}\right) \varphi_{Y_2}\!\left(\frac{t}{\sqrt{n}}\right) \cdots \varphi_{Y_n}\!\left(\frac{t}{\sqrt{n}}\right) \ = \ \left[\varphi_{Y_1}\!\left(\frac{t}{\sqrt{n}}\right)\right]^n,$$

where in the last step we used the fact that all of the Y_i are identically distributed. The characteristic function of Y_1 is, by Taylor's theorem,

$$\varphi_{Y_1}\!\left(\frac{t}{\sqrt{n}}\right) \ = \ 1 - \frac{t^2}{2n} + o\!\left(\frac{t^2}{n}\right), \quad \left(\frac{t}{\sqrt{n}}\right) \to 0$$

where $o(t^2)$ is little o notation for some function of t that goes to zero more rapidly than t^2. By the limit of the exponential function $\left(e^x = \left(1 + \frac{x}{n}\right)^n\right)$, the characteristic function of Z_n equals

$$\varphi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n \to e^{-\frac{1}{2}t^2}, \quad n \to \infty.$$

All of the higher order terms vanish in the limit $n \to \infty$. The right hand side equals the characteristic function of a standard normal distribution $N(0,1)$, which implies through Lévy's continuity theorem that the distribution of Z_n will approach $N(0,1)$ as $n \to \infty$. Therefore, the sum $X_1 + \cdots + X_n$ will approach that of the normal distribution $N(n\mu, n\sigma^2)$, and the sample average

$$S_n = \frac{X_1 + \cdots + X_n}{n}$$

converges to the normal distribution, $N\left(\mu, \frac{\sigma^2}{n}\right)$. □